

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Code of Conduct:

*I Arya. B. Nair - 192424133 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Arya

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided

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Answers

1. Given:

- **Training MSE = 0.85**
- **Testing MSE = 5.20**

Since the testing MSE is much higher than the training MSE, the model is **overfitting**.

Possible Reasons

- **Overfitting:** The model has memorized the training data instead of learning general patterns.
- **Insufficient Training Data:** A small dataset may not represent all possible cases.
- **Model Too Complex:** Too many layers or neurons can cause the network to fit noise in the training data.
- **Noise or Outliers:** Incorrect or unusual data points can increase testing error.
- **Improper Data Split:** Training and testing data may have different distributions.

Corrective Measures

- **Increase the size of the training dataset.**
- **Use regularization techniques** such as L1/L2 regularization or Dropout.
- **Reduce model complexity** by decreasing the number of layers or neurons.
- **Apply Early Stopping** to prevent over-training.
- **Perform proper data preprocessing** (remove noise, normalize data).
- **Use Cross-Validation** to improve the model's ability to generalize.
- **Tune hyperparameters** such as learning rate, batch size, and number of epochs.

Conclusion

Since **Testing MSE (5.20) > Training MSE (0.85)**, the model suffers from overfitting. Applying regularization techniques and improving data quality can reduce the testing error and improve generalization.

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(10-1)

Name : Arya . B . Nair

Reg No : 192424133

Course : MCA04 - Deep Learning

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyze the possible reasons for this difference and suggest suitable corrective measures.

Training MSE = 0.85

Testing MSE = 5.20

→ Model exhibits overfitting because the training error is significantly ~~error~~ lower than the testing error. The model has learned the training data well but fails to generalize to unseen data.

POSSIBLE REASONS :-

- Overfitting due to excessive model ^{complexity} $\wedge$.
- Insufficient training data well but fails to generalize to unseen data.
- Presence of ~~noisy~~ or irrelevant features.
- Inadequate regularization.
- Distribution mismatch between training and testing datasets.

COLLECTIVE MEASURES

- Increase the size of the ~~training~~ training dataset.
- Apply $L1/L2$ regularization or dropout.
- Reduce model complexity.
- Use early stopping.
- Perform cross-validation.
- Tune hyperparameters such as learning rate, epochs, and batch size.

Conclusion —

The model is overfitting. Regularization, more data, and simpler models can improve testing performance.